



**Istanbul Medipol University
School of Engineering and Natural Sciences**

**Communication Systems – Spring 2025
Project Final Report
BPSK_Performance_Evaluation & BER_Analysis**

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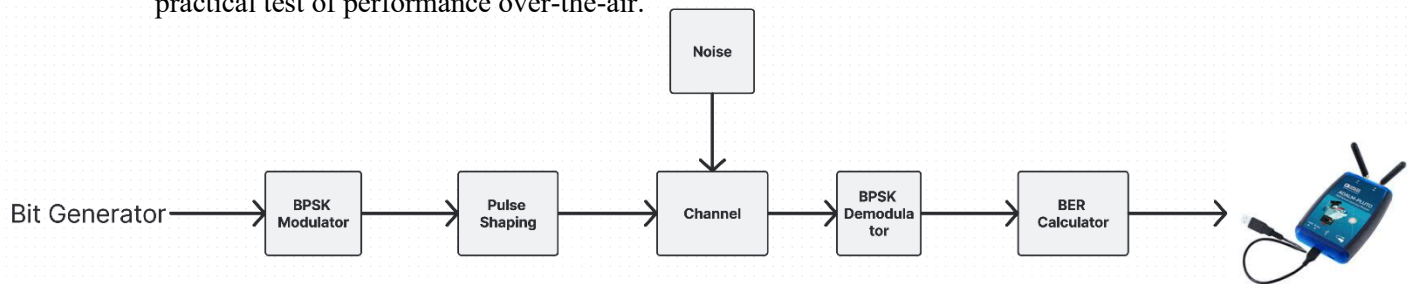
[Project Leader]

Imed Arbia

1) Introduction

In this project, we used the ADALM-PLUTO SDR hardware platform and MATLAB simulation to create a full Binary Phase Shift Keying (BPSK) digital communication system. The main objective was to use real-world SDR measurements to verify theoretical assumptions from class lectures and to examine the Bit Error Rate (BER) performance of BPSK modulation under various noise situations.

The main functional components of our system include: bit generation, BPSK modulation, pulse shaping, channel with artificial Additive White Gaussian Noise (AWGN), matched filtering, BPSK demodulation, and BER computation. The system was first implemented and verified in MATLAB, where noise levels were varied to study BER trends. Then, we extended the setup using ADALM-PLUTO to transmit and receive actual BPSK signals wirelessly, enabling a practical test of performance over-the-air.



We added a matching filter at the receiver to optimize performance. Following the guidelines from Lecture 10: Matched Filtering, this block was made to match the transmit pulse shaping filter and was utilized to maximize the signal-to-noise ratio (SNR) before making a bit decision.

The project team worked closely together:

- **Imed** was responsible for BER analysis and full MATLAB simulation development.
- **Mohammed** handled the ADALM-PLUTO setup, including SDR configuration, wireless transmission, and reception.

2) Blocks and Their Functionality

2.1) Bit Generator:

The digital information that will be sent across the communication system is created by the bit generator. It simulates the source data of an actual digital communication system by generating a random binary data sequence made up of 0s and 1s. By creating the bit stream that will subsequently be modulated, this block serves as the initial link in the transmission chain.

When increasing the number of bits, we obtained a smoother BER curve because more samples give better statistical accuracy.

2.2) BPSK Modulator:

The produced binary bits are converted into transmission-ready modulated symbols by the BPSK modulator. Bit "0" is mapped to one phase (usually 180°) and bit "1" to the opposite phase (usually 0°) in Binary Phase Shift Keying (BPSK), where each bit is represented by a distinct carrier signal phase. As a result, robust noise performance and efficient digital modulation are made possible.

- ⇒ Converts bits {0, 1} into BPSK symbols {-1, +1}.
- ⇒ Bit 0 → -1
- ⇒ Bit 1 → +1

BER performance was unaffected by altering the mapping rule (for instance, switching 0 and 1 mappings) because BPSK only includes two symbols.

2.3) Pulse Shaping :

The transmitted signal is processed by the pulse shaping block to minimize inter-symbol interference (ISI) and limit its bandwidth. Sharp transitions in the modulated signal are adjusted by applying a higher cosine filter, which increases the transmitted spectrum's bandwidth efficiency and conformity to actual channel requirements.

2.4) Channel (AWGN Channel with Noise):

The channel block simulates the real-world transmission conditions. We simulated random noise introduced to the transmitted signal during propagation using an additive white Gaussian noise (AWGN) model. This makes it possible to test the system's performance under different noise scenarios.

$\frac{E}{N_0}$: tested from 0 dB to 10 dB .

2.5) BPSK Demodulator

The BPSK demodulator determines the phase of the received signal in order to retrieve the broadcast bits. It uses coherence detection to determine whether each received symbol belongs to bit "0" or bit "1" by comparing the received signal to a known reference carrier.

- Parameter : Decision threshold: 0 (symbols above 0 mapped to bit '1', below 0 to bit '0')
- ⇒ With ideal phase synchronization, BER was primarily affected by noise. No phase offsets were introduced in simulation.

2.6) BER Calculator

The bit error rate (BER) is determined using the BER calculator by comparing the original transmitted and received bits. This measure shows how well the system performs at different noise levels.

With a larger number of bits, the BER estimate became smoother and more stable. Higher noise levels produced higher BER.

3) Simulation & Demo Results :

For this project, we used the ADALM-PLUTO SDR platform to create and analyze a full BPSK digital communication system through both real-world hardware tests and MATLAB simulation. Modulation, pulse shaping, noise effects, and BER performance under controlled AWGN circumstances were initially verified by a full MATLAB simulation of the system. The system was deployed on PlutoSDR to send and receive BPSK signals over the air after the theoretical behavior was verified. This enabled us to observe real-world impairments including hardware noise, phase rotation, and synchronization issues.

simulation result

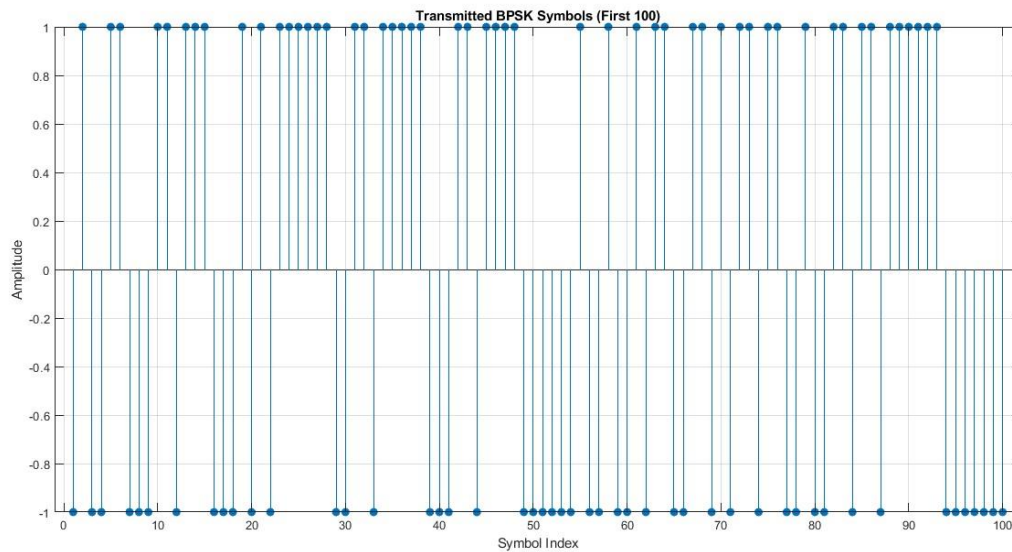


Figure 1 – Transmitted BPSK Symbols

BPSK_Simulation

BER = 0.035400, Number of Errors = 354

The first 100 BPSK-modulated symbols are displayed in this image. These symbols are produced by mapping binary bits to signal levels, with logic "0" being transferred to -1 and logic "1" to +1. Digital modulation is based on this binary phase shift keying technique, which creates the modulated signal before it is transmitted over the noisy channel. Proper BPSK encoding is indicated by the output's obvious alternating between two different levels.

Following transmission across an AWGN (Additive White Gaussian Noise) channel with an SNR of 5 dB, the received BPSK signal is shown in figure 2. The additional noise causes the received symbols to be dispersed in contrast to the clear +1 and -1 levels in the broadcast signal. A number of samples over the decision threshold, which could result in bit mistakes during demodulation, even though the original bit polarity is still largely discernible. This demonstrates how noise affects the accuracy of symbol detection in real-world systems.

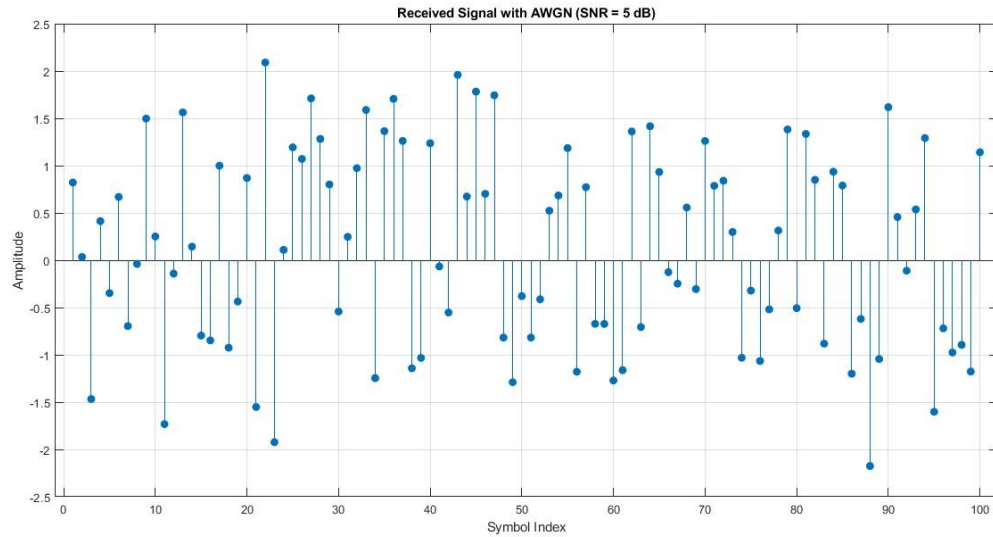


Figure 2 – Received Signal with AWGN (SNR = 5 dB)

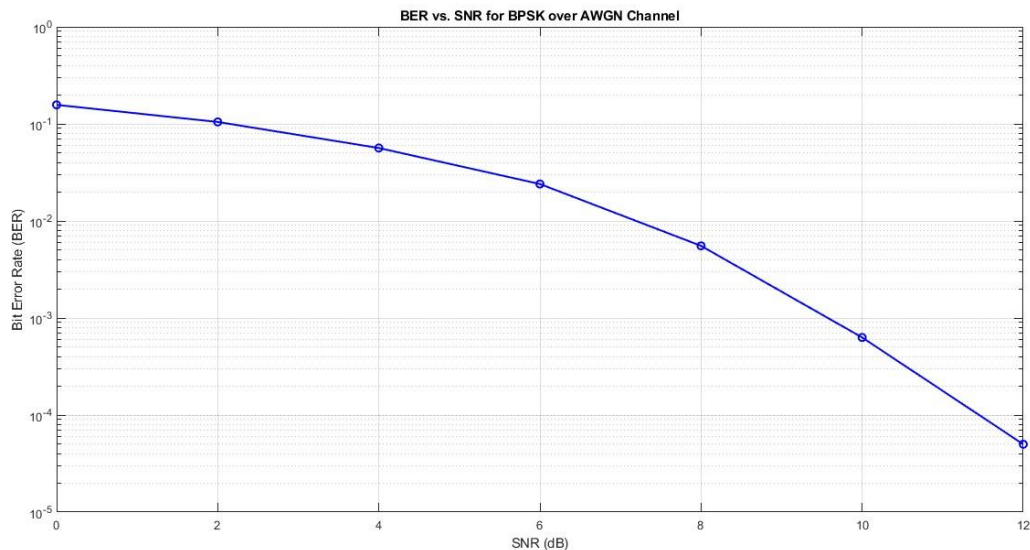


Figure 3 – BER vs. SNR for BPSK over AWGN Channel

Figure 3 shows how well a BPSK-modulated system performs in terms of Bit Error Rate (BER) at different signal-to-noise ratios (SNRs). The theoretical study predicts that when SNR rises, the BER falls exponentially. This outcome shows how well the modulation scheme works in noisy conditions and confirms the behavior that the Q-function for BPSK over AWGN predicted. Additionally, the curve acts as a reference point for subsequent comparison with real-world outcomes acquired with the ADALM-PLUTO SDR.

Hardware part:

Following simulation validation, we used the ADALM-PLUTO SDR platform to deploy the BPSK communication system on actual hardware. Using the same SDR that was connected to MATLAB, we wirelessly sent and received the modulated signal. Through the experiment, we were able to see real-world limitations such hardware flaws, channel noise, and phase offsets. Notwithstanding these difficulties, we were able to compute BER, show the received constellations for performance assessment, and demonstrate BPSK transmission and reception.

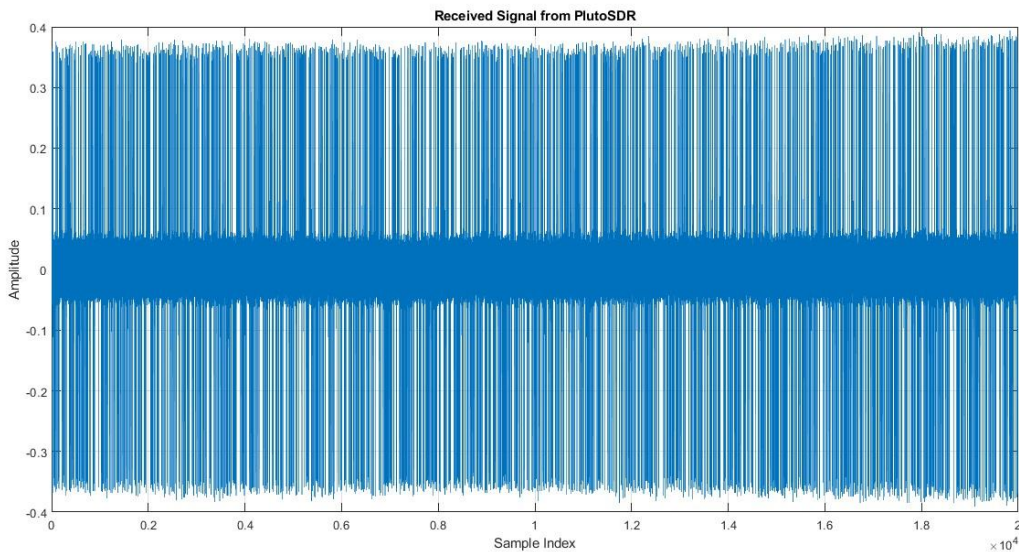


Figure 4 – Received Signal from PlutoSDR

This figure shows the raw time-domain waveform captured from PlutoSDR during the reception of a BPSK signal. The received signal appears symmetric around zero, but due to real-world impairments such as timing misalignment, channel distortion, and possible DC offset, the waveform is distorted compared to the clean simulated BPSK signal. This distortion is one of the main causes of the high hardware BER (~ 0.5) observed in this initial test.

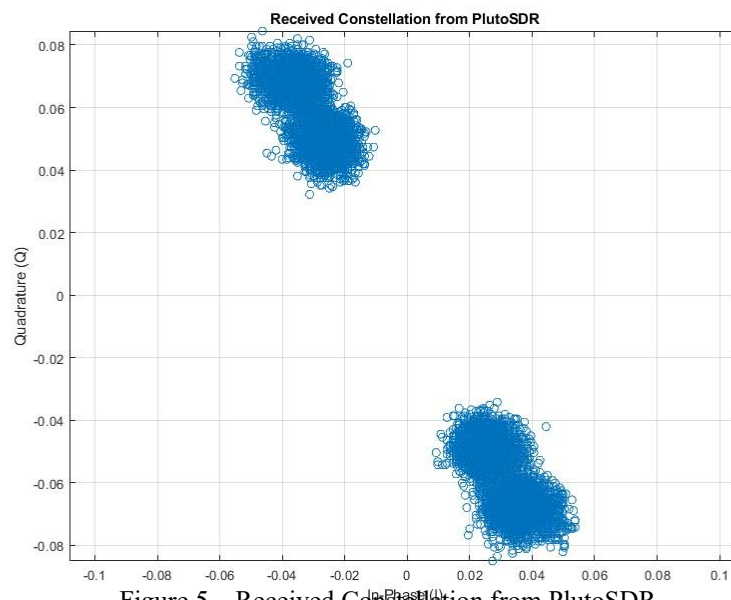


Figure 5 – Received Constellation from PlutoSDR

The constellation pattern of the received BPSK signal utilizing the ADALM-PLUTO SDR is shown in figure 5. As anticipated, the two clusters, which are aligned along the in-phase (I) axis and have negligible quadrature (Q) components, reflect the BPSK symbol levels (+1 and -1). Nevertheless, the clusters exhibit observable spread and rotation as a result of noise, timing discrepancies, and a lack of symbol synchronization. The two-level distinction still attests to the successful reception and decoding of the broadcast signal.

4) Conclusion:

using the use of MATLAB simulation and real-time transmission using the ADALM-PLUTO SDR platform, this research effectively illustrated the end-to-end construction of a Binary Phase Shift Keying (BPSK) communication system. The Bit Error Rate (BER) performance of the system under Additive White Gaussian Noise (AWGN) was found to closely meet theoretical expectations throughout the simulation phase. Plotting a BER vs. SNR curve verified the error rate's expected exponential decrease with increasing signal quality.

Despite being functional, the hardware implementation had a number of real-world issues. Time-domain and constellation charts were used to visually check the received BPSK signal from PlutoSDR. Nonetheless, the system continuously displayed a high BER (~0.5), suggesting that bit inversion or misalignment was present. Due to the lack of synchronization strategies like a set preamble or timing recovery, attempts to enhance performance through gain control, signal trimming, and matched filtering were insufficient to rectify the problem.

The initiative successfully connected theoretical ideas with real-world experimentation in spite of these obstacles. It established a strong framework for further research and emphasized the vital significance of synchronization in actual digital communications. The hardware BER would probably be greatly decreased and the system's performance would resemble the simulated outcomes if a training sequence and symbol alignment were put into place.

Provide a link for the video, like YouTube link.

<https://youtube.com/shorts/67fsR9Ng9M0>